PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : THERMODYANAMIC & THERMOCHEMISTRY (Question Bank # 04)

SINGLE CORRECT

1. **(C)**

$\boxed{n_{He}=1.2}$ $\boxed{n_{He}=1.2}$

vessel-1 vessel-2

$v_2=2v_1$ & $T_2=\frac{1}{2}T_1$

To find s_2-s_1 we may think that gas expanded from vessel-1 to vessel-2 condition so

$$\Delta s = nc_v \ell n\frac{T_2}{T_1}+nR\ell n\frac{v_2}{v_1}$$

$$=n\times\frac{3}{2}R\times\ell n\frac{1}{2}+nR\ell n2$$

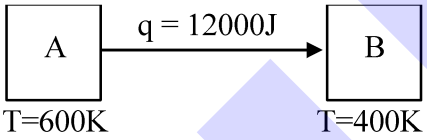
$$=-n\times 1.5\,R\ell n\,2+nR\ell n2$$

$$=-0.5\times n\times R\times \ell n\,2$$

$$=-0.5\times 1.2\times 2\times 0.7$$

$$=-0.84\ \text{cal/k}$$

2. **(B)**



$\Delta S_A=-\frac{12000}{600}$ As heat is lost

$=-20$ J/K

$\Delta S_B=\frac{+12000}{400}=+30$ J/K As heat is gained

$\Delta S_{Total}=-20+30=+10$ J/K

3. **(C)**

$2NO_2(g)\rightleftharpoons N_2O_4(g)$

$\Delta U^o_{rxn}=\Delta U^o_f N_2O_4(g)-2\times\Delta U^o_f NO_2(g)$

$-16=2-2\times\Delta U^o_f NO_2(g)$

$\Delta U^o_f NO_2(g)=9\text{kcal/mole}$

For formation of $NO_2(g)$

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$\frac{1}{2}N_2(g) + O_2(g) = NO_2(g)$

$\Delta n_g = (1) - (1 + \frac{1}{2}) = -\frac{1}{2}$

&

$\Delta H_f^\circ NO_2(g) = \Delta U_f^\circ + \Delta n_g RT$

$= (9 - 1) = 8$ kcal/mole

4. **(B)**

$PV^3 = \text{constant}$ so polytropic constant = 3

For ideal gas $v = \frac{nRT}{P}$

so $P \times \frac{n^3R^3T^3}{P^3} = \text{constant}$

or $P^{-2} \times T^3 = \text{constant}$

$P_1 = 1 \quad T_1 = 300$ | $1 \times (300)^3 = 1/8 \times T_2^3$

$P_2 = 2\sqrt{2} \quad T_2 = ?$ | $T_2 = 600$ K

now for polytropic process

$w = \frac{nR\Delta T}{x - 1}$

$w = \frac{1 \times R \times 150 \times 2}{2}$

$= 150$ R

5. **(C)**

$\Delta G^\circ = -RT\ell nK_p = \Delta H^\circ - T\Delta S^\circ$

$\ell nK_p = -\frac{\Delta H^\circ}{R}\frac{1}{T} + \frac{\Delta S^\circ}{R}$

$\log K_p = -\frac{\Delta H^\circ}{2.303R} \times \frac{1}{T} + \frac{\Delta S^\circ}{R}$

$\text{slope} = -\frac{\Delta H^0}{2.303R} = -1$

$\Delta H^0 = 2.303\ R = 4.606$ cal/mole

6. **(B)**

$S_{gas}^\circ > S_{liq}^\circ > S_{gas}^\circ$

- So A & C are correct
- $(\Delta G < 0)_{T,P}$ at constant temperature & press is condition for spontaneous reaction
- A spontaneous reaction is may be eitehr SLOW or FAST

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7. **(B)**

$q_{Rev} = -1200$ cal at 300 k

$\Delta S_{gas} = -\frac{1200}{300}$ cal / kelvin

For isothermal $\Delta H = 0$

So $\Delta G = \Delta H - T\Delta S$

$= -T\Delta S$

$= -T \times \frac{-1200}{300} = +1200$ cal

8. **(A)**

$H^+_{(aq)} + HO^-_{(aq)} = H_2O_{(\ell)}$

$\Delta H^\circ_{rxn} = \Delta H^\circ_{Neutralisation}$

$= \Delta H^\circ_f H_2O_{(\ell)} - \Delta H^\circ_f H^+_{(aq)} - \Delta H^\circ_f \left(HO^- aq\right)$

& $\Delta H^\circ_f \left(H^+\right)_{aq} = 0$

-57.3 kJ/mole $= -285.8$ kJ/mole $- \Delta H^\circ_f HO^-_{(aq)}$

$\Delta H^\circ_f \left(\overline{O}H\right)_{aq} = -228.5$ kJ/mole

9. **(C)**

OA $\rightarrow$ heating of solid

AB $\rightarrow$ meltingof solid

BC $\rightarrow$ heating of liquid

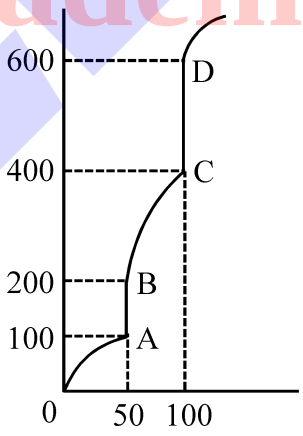
CD $\rightarrow$ vaporisation of liquid

For CD $\Delta S_{vap} = 200$ Joule

$= \frac{\Delta H_{vap}}{T}$

$\Delta H = 200 \times 100$ Joule

$\Delta H_{vap} = 20$ kJ



MULTIPLE CORRECT

10. **(B,D)**

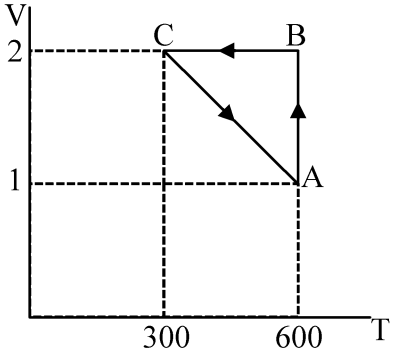
(A)	Isothermal expansion	T = constant $E_1 = 0$	so no change in K.E.
(B)	Adiabatic expansion	Temp full	$E_2 = -$ ve
(C)	Free expansion	T = constant	$E_3 = 0$
(D)	Isobaric expansion T increase so $E_4 > 0$ Hence, $E_1 = E_3 = 0$ and $E_4 > E_3$	$V \propto T$	

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11. (A,C)

1 mole $N_2(g)$ $\quad C_V = \frac{5}{2}R \qquad C_P = \frac{7}{2}R$



$AB \rightarrow$ Isothermal

$W_{AB} = -nRT\ell n\, v_2/v_1$

$= -1 \times 2 \times 600 \times \ell n2$

$= -840$ cal

$\Delta U_{CA} = n \times c_v \times \Delta T$

$= 1 \times \frac{5}{2}R \times 300$

$= 1500$ cal

so (A) & (C) correct.

$BC \rightarrow$ Isochoric

$\Delta H_{AB} = 0$

$\Delta U_{AB} = 0$

BC w = 0

$\Delta U_{BC} = q_{BC}$

$nc_v\Delta T = q_{BC}$

$q_{BC} = -1500$ cal

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12. (B,D)

(1) $MX_2(s) = M^{2+}_{(g)} + 2X^-_{(g)}$ $\quad \Delta H = +800$ kJ/mole

(2) $MX_2(s) + (aq) = M^{2+}_{(aq)} + 2X^-_{(aq)}$ $\quad \Delta H = 50$ kJ/mole

(3) $M^{2+}_{(g)} + (aq) = M^{2+}_{(aq)}$ $\quad \Delta H_{HYD}M^{2+}_{(g)} = x$

(4) $X^-_{(g)} + (aq) = X^-_{(aq)}$ $\quad \Delta H_{HYD}X^-_{(g)} = y$

equation (2) – equation (1)

$\Rightarrow M^{2+}_{(g)} + 2X^-_{(g)} + (aq) = M^{2+}_{(aq)} + 2X^-_{(aq)}$

$\Delta H = -750$ kJ/mole

equation (3) + 2 × equation (4)

$\Rightarrow M^{2+}_{(g)} + 2X^-_{(g)} + aq = M^{2+}_{(aq)} + 2X^-_{(aq)}$

$= x + 2y$

so $x + 2y = -750$

& $\frac{x}{y} = \frac{3}{1}$ $\quad$ so $\quad 5y = -750$

$y = -150$ kJ/mole

$x = -450$ kJ/mole

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13.(B); 14.(C); 15.(D)

(I) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H = -ve$ combustion reaction
$\Delta n_g = 0$

(II) $SO_3(g) \rightarrow SO_2(g) + 1/2\ O_2(g)$ $\Delta H = +ve$ Reverse of combustion of $SO_2(g)$
$\Delta n_g = +1/2$

(III) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$ $\Delta H = -ve$ combustion reaction
$\Delta n_g = -1$

(IV) $C(s) + 1/2\ O_2(g) \rightarrow CO(g)$ $\Delta H = -ve$ combustion reaction
$\Delta n_g = -1/2$

INTEGER

1. (100.00)

$\Delta H = \Delta U + P\Delta V$

$|\Delta H - \Delta U| = |P\Delta V|$

For 1 mole graphite = 12 gm graphite

$$V_{graphite} = \frac{12}{2} cm^3 = 6ml$$

For 1 mole Diamond = 12 gm diamond

$$V_{Diamond} = \frac{12}{3} = 4ml$$

Graphite $\rightarrow$ Diamond

$\Delta V = V_D - V_C$

$= -2$ ml

$|\Delta H - \Delta V| = 500 \times 10^3$ bar $\times (-2) \times 10^{-3}$ Litre

$= 1000$ bar litre

$= 1000 \times 100$ joules

$= 100$kJ/mole

2. (4.00)

n moles of compound $= \frac{8.4}{8 \times 10^4}$

$q = 5 \times v \times \Delta T$ as 's' is in per k per ml

$= 4.2 \times 25 \times 0.04$

$= 4.2$ joules

for $\frac{8.4}{8 \times 10^4}$ moles of compound q = 4.2 joules

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for 1 mole compound $= \frac{4.2\times8\times10^4}{8.4}$

10 mole oxygen = 4×10^4 joule

for 1 mole oxygen q = 4×10^3 g

= 4kJ

3. **(5.00)**

dG = VdP – SdT

T = constant

dG = VdP

$\int_{G_1}^{G_2} dG = v\int_{P_1}^{P_2} dP$

$\Delta G = v(P_2 - P_1)$

$\Delta G = \frac{13}{0.78} cm^3 \times 3000 \text{ bar}$

= 5kJ

4. **(6.00)**

For Rev. Ism process $q_{rev} = w$

$\Delta S = \frac{q_{rev}}{T}$

W = T × ΔS

= 300 × 20 = 6 kJ

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MATCHING CORRECT

1. **(A)-P, T ; (B)-P,Q,R,T ; (C)-P,Q,R,S ; (D)-P,R,T**

(A) $H_2O(\ell) \rightleftharpoons H_2O(g)$ 100°C 1atm

ΔG = 0 w = (–ve) q > 0 ΔS > 0 ΔH > 0

(B) Isothermal process ΔH = 0 ΔU = 0

expansion ΔS > 0 q = (–w)

so ΔG < 0 so q > 0

as (w = – ve)

(C) AdB free expansion

W = 0 ΔU = 0 q = 0 ΔH = 0 ΔS > 0

ΔG < 0

(D) Isobaric heating

ΔH > 0 W < 0 ΔS > 0 ΔG < 0

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